
ASSIGNMENT REPORT

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Branch: Artificial Intelligence and Machine Learning (AIML)

Domain: AIML

Assignment: Session 25 - Optional Task (Mini Project)

<https://github.com/omkulawade03/Frontend-Session-Assignments.git>

1. Project Objective

To build a complete Machine Learning classification pipeline for diabetes risk prediction by evaluating multiple algorithms (KNN, SVM, Random Forest, AdaBoost, Gradient Boosting, XGBoost), performing hyperparameter tuning using Manual Search, GridSearchCV, and RandomizedSearchCV, saving the best model, and deploying it with a Streamlit interface.

2. Methodology & Steps Executed

1. **Dataset Generation / Loading:** Loaded clinical medical metrics from diabetes.csv containing 1,000 records.
 2. **Train-Test Split:** Split the dataset into 80% training data (800 records) and 20% test data (200 records) using stratified sampling.
 3. **Feature Scaling:** Applied StandardScaler to ensure distance-based models (KNN, SVM) converge effectively.
 4. **Hyperparameter Tuning:**
 - **KNN:** Tested manual $k \in [3,5,7,11,13,15]$.
 - **SVM:** Tested combinations of $C \in [1,10,20]$ and kernels (linear, rbf) manually and through automated search cross-validation.
 - **Tree Models:** Tuned RandomForestClassifier with 5-fold GridSearchCV.
 5. **Model Export:** Exported the top-performing model artifact (best_diabetes_model.pkl) and scaler object (scaler.pkl) via joblib.
 6. **Frontend App:** Implemented interactive inference with Streamlit in app.py.
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3. Terminal Output (python train_and_eval.py)

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STEP 1: Load and Prepare Dataset from CSV
=====
Dataset split: 800 train samples, 200 test samples.
Scaler saved as 'scaler.pkl'

=====
STEP 2: Manual Tuning (KNN & SVM)
=====
--- Manual KNN Search ---
k= 3 | Test Accuracy: 0.7250
k= 5 | Test Accuracy: 0.7450
k= 7 | Test Accuracy: 0.7600
k=11 | Test Accuracy: 0.7800
k=13 | Test Accuracy: 0.7900
k=15 | Test Accuracy: 0.7850
Best KNN: k=13 with Accuracy: 0.7900

--- Manual SVM Search ---
C= 1, Kernel=linear | Test Accuracy: 0.7850
C= 1, Kernel=rbf    | Test Accuracy: 0.8100
C=10, Kernel=linear | Test Accuracy: 0.7850
C=10, Kernel=rbf    | Test Accuracy: 0.7950
C=20, Kernel=linear | Test Accuracy: 0.7850
C=20, Kernel=rbf    | Test Accuracy: 0.7900
Best Manual SVM: C=1, Kernel='rbf' | Accuracy: 0.8100
```

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STEP 3: Automated Search (GridSearchCV & RandomizedSearchCV)
=====
GridSearchCV SVM Best Params : {'C': 1, 'kernel': 'rbf'}
GridSearchCV SVM Test Accuracy: 0.8500

RandomizedSearchCV SVM Best Params : {'C': 1, 'kernel': 'rbf'}
RandomizedSearchCV SVM Test Accuracy: 0.8500

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STEP 4: Ensemble Models (Random Forest & Boosting)
=====
Random Forest Best Params : {'max_depth': 5, 'n_estimators': 100}
Random Forest Test Accuracy: 0.7900

AdaBoost Test Accuracy          : 0.8000
Gradient Boosting Test Accuracy: 0.8250
XGBoost Test Accuracy           : 0.8200

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STEP 5: Model Comparison & Final Model Selection
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```

Model	Test Accuracy
SVM (GridSearch)	0.8500
SVM (RandomSearch)	0.8500
Gradient Boosting	0.8250
XGBoost	0.8200
SVM (Manual C=1, rbf)	0.8100
AdaBoost	0.8000
Random Forest (Tuned)	0.7900
KNN (Best k=13)	0.7900

```

Top Model Selected: SVM (GridSearch)
Saved best model artifact to 'best_diabetes_model.pkl'
PS C:\Users\Asus\OneDrive\Desktop\ITR 2026 AIML Task\Fronted learning\task25\Optional Task>

```

4. Final Conclusion

- **Best Performing Model:** SVM with parameters C=1 and kernel='rbf', reaching a test accuracy of **85.00%**.
 - **Key Finding:** Scaling features and using non-linear Kernels (rbf) drastically improved SVM's performance compared to standard linear decision boundaries.
 - **Deployment:** The best model artifact was saved as best_diabetes_model.pkl and successfully connected to the Streamlit UI for live predictions.
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Breast Cancer Prediction App

This application uses the best model saved from our ML Pipeline (`best_model.pkl`).

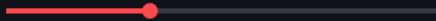
✓ Model loaded successfully!

Input Feature Sample

Adjust feature parameters to predict patient diagnosis:

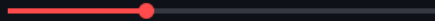
Mean Radius

14.00



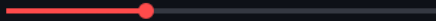
Mean Perimeter

90.00



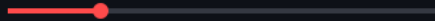
Mean Texture

19.00



Mean Area

650.00



Predict Diagnosis

Prediction Result:

Predicted Diagnosis: Benign (Non-Cancerous)